THE UNIVERSITY OF TEXAS AT DALLAS

CS4485.0w1 – Computer Science Project

Fall 2026

Project Weekly Report

Project Title	Air Pollution Analyzer and Predictor
Faculty Advisor	Muhammad Ikram
Group Number (E.g., Ikram_T1_30)	Ikram_T7_36
Student Names	Chris Abellana, Aldo Carmona, Alexander Constante, Andy Do, Faiz Khan, Yael Rico
Group/Team Lead	Alexander Constante
Reporting Week (E.g., 9/14/26 to 9/18/26)	10/19/2026 to 10/23/2026
Meeting Attendees (students, faculty advisor, TA, etc.)	Students

Note. Submit completed report before next meeting. Add more bullets if needed.

1. Progress made in the reporting week (list all tasks who worked on it, and time spent on each task):
    a. Finalized training XGBoost model with rest of aqi baseline data (Yael 30 mins)
    b. Trained Random Forest Regression and LightGBM with the rest of baseline data (Yael 10 mins)
    c. Created a basic front end using React - Aldo (1hr 30 mins)
    d. Added disclaimer section to front end - Alex (20 mins)
    e. Data research, re-evaluation, and gathering - Andy (2hr 25 mins)
    f. Model comparisons - Andy (10 mins)

2. Challenges faced in the reporting week:
    a. Baseline data provided low accuracy metrics
    b. Other commitments (Classes, Jobs, etc.)

3. Plans for next week:
    a. Improve correlation of model
    b. Find errors with current data and fix
    c. Find additional features and test accuracy with them